☀️ Yuma Thermal Safety Protocol (SSoT)

Single Source of Truth (SSoT) for EXW265: Exercise for Special Populations Institutional Authority: Arizona Western College / EXW Master Library

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🏛️ 1. Mission Intent

In the Yuma, Arizona environment, ambient temperatures frequently exceed **110°F (43°C)**. For clinical populations, particularly those with **Multiple Sclerosis (MS)** or **Cardiovascular Disease**, physical accessibility (ADA) is insufficient without **Thermal Integrity**. This protocol defines the architectural and physiological requirements for safe facility ingress and exercise.

🛡️ 2. The Clinical Redline: Uhthoff's Phenomenon

For clients with MS, a core temperature increase of as little as **0.5°C** can trigger **Uhthoff's Phenomenon**—a temporary exacerbation of neurological symptoms due to nerve conduction failure in demyelinated axons.

📡 Thermal Audit Thresholds

- **Safe Zone:** Ambient temp < 85°F.
- **Caution Zone:** 86°F – 104°F. Requires "Thermal Shadow" for all ingress.
- **Redline Zone:** > 105°F. **Structural Failure** if no cooling vestibule or shaded path exists.

🏗️ 3. Architectural Standards (The Audit)

3.1: Ingress & The "Thermal Shadow"

Any path of travel from ADA parking to the facility entrance must be covered by a continuous canopy if the distance exceeds **30 feet**.

- **Asphalt Risk:** Unshaded black asphalt can reach surface temperatures of **160°F+**, posing a direct burn risk for mobility device users and increasing radiant heat gain.
- **Requirement:** Mandatory shade structures or heat-reflective "Cool Pavement" coatings.

3.2: The Cooling Vestibule (Air-Lock)

Facilities must utilize a **double-door vestibule system**.

- **Logic:** A single-door entry allows a "Thermal Bleed" where 115°F air enters the climate-controlled exercise zone.
- **Staging:** The vestibule serves as a "Pre-Cooling Zone" where clients can stabilize before beginning physical activity.

3.3: Solar Gain & Fenestration

Exercise equipment (Cardio/Strength) must not be placed within **5 feet** of unshaded South or West-facing windows.

- **Standard:** Windows must have **UV-3000** or higher thermal tinting to reduce radiant load.
- **Shadow Rule:** If no tinting exists, equipment must be moved to "Cool Zones" (interior walls).

🧪 4. Quantitative Scorecard (Review)

Use these points to generate the **Thermal Integrity Score** for any facility:

1. **Continuous Shade Path:** +3 Points
2. **Double-Door Vestibule:** +3 Points
3. **Hydration < 15ft from Entry:** +2 Points
4. **UV Tinting/Solar Control:** +2 Points

Threshold: 8/10 is required for "Clear for Takeoff" status.

🏁 5. Institutional Compliance

Any deviation from these standards during a **Clinical Audit (Module 10.3)** must be documented as a **Structural Failure** in the Pilot's Flight Log.